MILITARY SPECIFICATION

CASE, CARTRIDGE, 40MM, XM195

PART for

1. SCOPE

1.1 This specification covers the metal part for one type of cartridge case designated as Case, Cartridge, 40MM, XM195.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-A-8625	-	Arodic Coatings, For Aluminum and Aluminum Alloys
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of

STANDARDS

FEDERAL

FED-STD-151	-	Metals; Test Methods

FSC: 1310

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MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

11-5-86

DRAWINGS

US ARMY MUNITIONS COMMAND

9207989	-	Cartridge Case, 40MM, XM195

PUBLICATIONS

ARMY

ET-9207989	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications and M118 cases that have met the requirements of specification MIL-C-60985 (See 4.3.2.2).

3.2 Part.-The part shall comply with all requirements specified on Drawing (Dwg.) 9207989 all associated drawings, and with all requirements specified in applicable specifications.

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3.3 Hydrostatic pressure.-The cartridge case, less vent holes, shall be capable of withstanding a hydrostatic pressure of 48,000 pounds per square inch (psi), minimum (min), without evidence of a rupture of the bottom of the charge cup cavity when tested as specified in 4.4.2. Cases especially prepared for this test shall be identified from regular cases by painting a red strip around the open end, or any positive identification approved by the procuring activity. This requirement does not apply if the alternative construction is applied.

3.4 Hardness.-The hardness of the area near the mouth and on the base of the cartridge case shall be 67 + 7 on the Rockwell 30 T Scale when tested as specified as in 4.4.4.

3.5 Workmanship.-All parts shall be fabricated and finished in a through workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning methods shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

3.6 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

3.7 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (see 6.4)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of the following:

a. One hundred thirty-five (135) cartridge cases.
b. One hundred (100) cartridge cases less vent holes (when applicable).
c. Four (4) test panels and/or parts of case, cartridge for salt spray.

All parts submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular

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production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Components will be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.2.2.1 Quantities for first article inspection.-After classification of defects and other inspections have been performed as specified in 4.2.2, the following quantities shall be subjected to the indicated tests:

4.2.2.1.1 Hydrostatic test of cartridge case.-Each of the one hundred (100) cartridge cases less vent holes shall be subjected to this test. This test does not apply if the alternative construction is applied.

4.2.2.1.2 Hardness.-One hundred (100) cartridge cases shall be subjected to this test.

4.2.2.1.3 Salt spray.-The four (4) test panels and/or parts of case, cartridge shall be subjected to this test.

4.2.2.1.4 Diameter of vent holes.-Twenty-five (25) cartridge cases shall be sectioned and the diameter of each vent hole measured by any method satisfactory to the Contracting Officer. This test does not apply if the alternative construction is applied.

4.2.3 Rejection.-If any component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of the rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first

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article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Case, cartridge (See Dwg. 9207989).-

Categories	Defects	Method of Inspection	Code No (See 6.2)
Critical			
1.	Three (3) or more vent holes blocked or missing	AG	01001
Major	100% Inspection		
101.	One (1) or two (2) vent holes blocked or missing	AG	01002
Major:	AQL 2.50 Percent		
102.	Hardness	Test (See 4.4.4)	01003
103.	Concentricity of largest outside diameter with second largest outside diameter	Gage	01004
104.	Concentricity of largest inside diameter with second largest outside diameter	Gage	01005
105.	Concentricity of base plug cavity with second largest outside diameter	Gage	01006
106.	Concentricity of powder charge cup cavity with base plug cavity	Gage	01007
107.	Diameter of powder charge cup cavity, min.	Gage	01008
108.	Diameter of base plug cavity	Gage	01009
109.	Depth from base to bottom of powder charge cup cavity	Gage	01010
110.	Second largest outside diameter (See 6.5)	Gage	01011
111.	Diameter of node	Gage	01012
112.	Parallelism of open end with base	Gage	01013
113.	Perpendicularity of base to second largest outside diameter	Gage	01014
114.	Diameter of vent holes (see 4.3.3.3)	Gage	01015
115.	Largest inside diameter (see 6.6)	Gage	01016
116.	Distance to flat of inside projection	Gage	01017
117.	Diameter of flange, max.	Gage	01018
118.	Total length	Gage	01019
119.	Burr in vent hole	Visual Equipment	01020

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120.	Unconfined second largest outside diameter	Gage	01021
121.	Depth of largest inside diameter, min	Gage	01022
122.	Depth of base plug cavity	Gage	01023
123.	Metal defective (See 6.7)	Visual	01024
Minor:	AQL 2.50 Percent		
201.	Thickness of flange	Gage	01025
202.	Length of base plug seating projection from bottom of base	Gage	01026
203.	Wall thickness, min.	Gage	01027
204.	Stamping misleading or unidentifiable	Visual	01028
205.	Protective coating with bare spot, except anode rack marks	Visual	01029
206.	Evidence of poor workmanship (See 3.5) except burrs	Visual	01030
207.	Color incorrect	Visual	01031

4.3.2.2 Case, cartridge Alternate construction (See Dwg. 9207989).-

Categories	Defects	Method of Inspection	Code No
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Concentricity of largest inside diameter with second largest outside diameter	Gage	02001
102.	Largest inside diameter (see 6.6)	Gage	02002
103.	Second largest outside diameter (see 6.5)	Gage	02003
104.	Parallelism of open end with base	Gage	02004
105.	Depth of largest inside diameter	Gage	02005
106.	Unconfined second largest outside diameter	Gage	02006
107.	Total length	Gage	02007
Minor:	AQL 1.50 Percent		
201.	Protective coating with bare spot except anode rack marks	Visual	02008
202.	Evidence of poor workmanship, except burrs (see 3.5)	Visual	02009
203.	Color incorrect	Visual	02010

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4.3.2.3 Box, fiber prior to sealing (See Dwg. 9223934).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Incorrect number of case cartridges	Visual	03001
102.	Insufficient fillers	Visual	03002
Minor:	AQL 1.00 Percent		
201.	Separator missing	Visual	03003
202.	Case cartridge inverted	Visual	03004

4.3.2.4 Box fiber packing (See Dwg. 9223934).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40		
101.	Container damaged	Visual	04001
Minor:	AQL 1.00 Percent		
201.	Contents loose	Manual	04002
202.	Label or marking missing or unidentifiable	Visual	04003

4.3.3 Testing.-

4.3.3.1 Salt spray (See Dwg. 9207989), Major defect, Code No. 05001.-Four (4) test panels and/or parts shall be selected from each lot and subjected to this test. Rejection shall be as specified in Specification MIL-A-8625. The test shall be conducted in accordance with 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.2 Process control inspection of bath temperature for anodic coating (See Dwg. 9207989), Major defect, Code No. 06001.- Continuous monitoring inspection shall be conducted for bath temperature requirement specified on the applicable drawing. If the bath temperature for a batch of parts does not comply with the applicable drawing requirement, the batch shall be suitably identified, classified defective and removed from the lot.

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4.3.3.3 Hydrostatic test of cartridge case (See Para. 3.3), Major defect, Code No. 07001.-Fifty (50) cartridge cases, less vent holes, shall be selected from each lot and subjected to this test. If any cartridge case fails to comply with the applicable requirement, the lot shall be rejected. The test shall be conducted in accordance with 4.4.2 using equipment in accordance with 4.3.4. This test does not apply if the alternative construction is applied.

4.3.3.4 Total vent hole area (alternate method) (See Dwg. 9207989), Major defect, Code No. 08001.-Each cartridge case shall be inspected for total vent hole area. If any cartridge case fails to comply with the requirement specified on the applicable drawing, it shall be classified defective and removed from the lot.

NOTE: If, upon calibration, the equipment setting has changed so as to permit acceptance of cartridge cases having total vent hole areas that deviate from the requirement specified on the applicable drawing, then the cases accepted since the last calibration must be reinspected and the defective parts removed from the lot. This test shall be conducted in accordance with 4.4.3 using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 9207989 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instruc ions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment drawings are needed.

4.4 Test methods and procedures.

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4.4.1 Salt spray.-The test panels and/or parts shall be tested as specifiedin Standard FED-STD-151, Method 811, (5 percent salt spray test). This test shall be conducted for 240 hours.

4.4.2 Hydrostatic test of cartridge case.-The sample cartridge case, less vent holes, shall be placed in a fixture as shown on the Equipment List. The protrusion inside of the case shall not be supported. The fixture shall be placed in a press and a load applied until the required pressure is reached. The ram shall be applied at a rate not exceeding 0.25 inch per minute. This test is a destructive test. Parts so tested shall not be returned to the lot. This test does not apply if the alternative construction is applied.

4.4.3 Total vent hole area (alternate method).-The cartridge case shall be inspected for total vent hole area using the equipment specified on the Equipment List. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.4 Hardness.-The test shall be conducted in accordance with Method 243 of Federal Standard No. 151. This is a non-destructive test. Parts so tested may be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.-

5.1.1 Level C -Minimum Military Package.-The case cartridges shall be packed for shipment in accordance with Dwg. 9223934.

5.2 Packing.-

5.2.1 Level C - Minimum Military Package.-The case cartridge packed for shipment as specified in 5.1.1 shall be packed in accordance with Dwg. 9223935.

5.3 Marking.-Marking of the cartons shall be in accordance with Dwg. 9223934.

5.4 Shipping.-When shipments of more than one lot are made on a carload, each lot must be kept separate and the dimension between lots clearly indicated to prevent mixing in transit.

6. NOTES

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6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of First Article Samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the number of drawings for Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.

MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army designs, the contractor shall comply

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with and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor-prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

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6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

6.5 If the requirements cannot be met in the free state, the "go" inside diameter plug gage may be inserted in the cartridge case while inspecting the "go" and "no go" second largest outside diameter. If the "go" inside diameter plug gage is utilized, the data cards shall be annotated accordingly.

6.6 If the requirement cannot be met in the free state, a restricted confining fixture may be utilized while inspecting the "go" and "no go" largest inside diameter. If the restricted fixture is utilized, the data cards shall be annotated accordingly.

6.7 Metal defective.-Nick, crack, dent or sharp edge which might interfere with the proper assembly or end item performance of the item.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1310-A-119

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